

Theoretical Derivations for a Lightweight Axisymmetric Taylor-Cone Electrospray Solver

ISEF Physics Project — Theory Deliverable

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Abstract

This document contains the theoretical derivations needed before constructing the computational model for an axisymmetric Taylor-cone / electrospray-onset solver. The goal is to define the continuous mathematical problem with enough precision that a separate implementation document can translate it into code. We derive the classical Taylor cone angle, identify the apex singularity, formulate axisymmetric Laplace and Poisson electrostatics, derive the Maxwell–Young–Laplace interfacial stress balance, define curvature in axisymmetric geometry, introduce effective space-charge shielding closures, and state the verification targets required before physical validation.

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1 Purpose of the theoretical model

The product we want is a lightweight solver that predicts Taylor-cone geometry and onset-like voltage trends for an electrospray emitter. The solver should be mathematically anchored in the known Taylor-cone limit and then extend that limit to include effective space-charge shielding near the cone apex.

The theory phase must answer the following questions before coding:

1. What is the ideal analytical solution we must reproduce?
2. What equations govern the electric potential in axisymmetric coordinates?
3. How does the electric field produce stress on the liquid surface?
4. How does surface tension enter through curvature?
5. How do we define a residual that measures whether a candidate interface is physically balanced?
6. How can space charge be introduced without claiming a full plasma/electrospray emission model?
7. What checks prove that the model is theoretically consistent before experiments begin?

The model is deliberately reduced-order. The first working product is not a full industrial multiphase CFD solver. It is an open, sparse, axisymmetric electrostatic-capillary solver with optional Poisson space-charge shielding.

2 Model hierarchy

We use a staged hierarchy:

Level	Model	Role
0	Classical Taylor cone	Analytical benchmark: recover the 49.3° cone half-angle.
1	Axisymmetric Laplace electrostatics	Solve charge-free potential and field.
2	Maxwell–Young–Laplace residual	Evaluate capillary/electric normal stress balance on an interface.
3	Free-boundary equilibrium	Adjust an interface shape until the residual is minimized.
4	Poisson space charge	Study how charge density shields the apex field.
5	Leaky dielectric / transport	Optional later extension, not required for the first solver product.

The first four levels define the core theoretical product.

3 Geometry and coordinates

We assume axisymmetry and use cylindrical coordinates

$$(r, \theta, z), \quad r \geq 0, \quad (1)$$

with no θ -dependence. The gas/air domain is Ω_g , the liquid domain is Ω_l , and the liquid–gas interface is Γ . The electric potential is ϕ , and

$$\mathbf{E} = -\nabla\phi. \quad (2)$$

An interface can be represented as a graph

$$r = R(z), \quad (3)$$

or parametrically as

$$s \mapsto (r(s), z(s)). \quad (4)$$

For the first model, graph form is sufficient except near possible overturning, which is not part of the initial onset calculation.

4 Classical Taylor cone derivation

4.1 Laplace equation outside a conducting cone

For an ideal perfectly conducting liquid in a charge-free gas, the gas-domain potential satisfies

$$\nabla^2\phi = 0. \quad (5)$$

Near the cone apex, spherical coordinates (ρ, ϑ) are natural, where ϑ is measured from the symmetry axis. The axisymmetric Laplace operator is

$$\nabla^2\phi = \frac{1}{\rho^2} \frac{\partial}{\partial\rho} \left(\rho^2 \frac{\partial\phi}{\partial\rho} \right) + \frac{1}{\rho^2 \sin\vartheta} \frac{\partial}{\partial\vartheta} \left(\sin\vartheta \frac{\partial\phi}{\partial\vartheta} \right). \quad (6)$$

A separated axisymmetric harmonic solution is

$$\phi(\rho, \vartheta) = A\rho^\nu P_\nu(\cos\vartheta), \quad (7)$$

where P_ν is a Legendre function of degree ν , not necessarily an integer.

4.2 Stress-scaling argument

The electric field scales as

$$E \sim |\nabla\phi| \sim \rho^{\nu-1}. \quad (8)$$

The electrostatic pressure scales as

$$p_E \sim \varepsilon_0 E^2 \sim \rho^{2\nu-2}. \quad (9)$$

For a cone, the curvature scale is $1/\rho$, so capillary pressure scales as

$$p_\gamma \sim \frac{\gamma}{\rho}. \quad (10)$$

Balancing electric pressure and capillary pressure requires

$$2\nu - 2 = -1, \quad (11)$$

after which

$$\nu = \frac{1}{2}. \quad (12)$$

Thus the local Taylor potential is

$$\phi(\rho, \vartheta) = A\rho^{1/2}P_{1/2}(\cos \vartheta). \quad (13)$$

4.3 Cone angle condition

A perfect conductor is an equipotential. If the cone surface is $\vartheta = \vartheta_0$, the angular part must vanish:

$$P_{1/2}(\cos \vartheta_0) = 0. \quad (14)$$

The relevant root is approximately

$$\vartheta_0 \approx 130.7^\circ. \quad (15)$$

The physical cone semi-vertical angle is therefore

$$\alpha_T = 180^\circ - \vartheta_0 \approx 49.3^\circ. \quad (16)$$

This is the most important analytical benchmark for the solver.

4.4 Apex singularity

With $\nu = 1/2$,

$$E \sim \rho^{-1/2}. \quad (17)$$

Therefore

$$E \rightarrow \infty \quad \text{as} \quad \rho \rightarrow 0. \quad (18)$$

This singularity is unphysical. In reality it is regularized by finite apex radius, charge emission, space charge, finite conductivity, viscosity, and molecular-scale effects. The computational model will regularize it through finite grid/interface resolution and through Poisson space-charge shielding.

5 Axisymmetric electrostatics

For an axisymmetric potential $\phi(r, z)$,

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2}. \quad (19)$$

5.1 Laplace model

The charge-free gas model is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (20)$$

5.2 Poisson model

With gas-domain charge density ρ_e ,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = -\frac{\rho_e(r, z)}{\varepsilon_0}. \quad (21)$$

5.3 Boundary conditions

The essential boundary conditions are:

$$\phi = V_0 \quad \text{on powered conductor or ideal liquid surface,} \quad (22)$$

$$\phi = 0 \quad \text{on grounded extractor electrode,} \quad (23)$$

$$\partial_r \phi = 0 \quad \text{on the symmetry axis } r = 0, \quad (24)$$

$$\partial_n \phi = 0 \text{ or } \phi = \phi_\infty \quad \text{on artificial far boundaries.} \quad (25)$$

The axis condition is required for regularity. The far-boundary condition is a numerical approximation and must be tested by moving the boundary farther away.

6 Maxwell stress and electric pressure

The electrostatic Maxwell stress tensor is

$$\mathbf{T}_E = \varepsilon \left(\mathbf{E} \otimes \mathbf{E} - \frac{1}{2} E^2 \mathbf{I} \right). \quad (26)$$

For a perfect conductor at electrostatic equilibrium, the electric field inside the liquid is zero and the tangential gas-side electric field vanishes on the surface. The normal electric pressure on the liquid surface reduces to

$$p_E = \frac{\varepsilon_g}{2} E_n^2, \quad E_n = \mathbf{E}_g \cdot \mathbf{n}. \quad (27)$$

For leaky dielectric extensions, the complete traction jump would use both gas and liquid Maxwell stresses:

$$\mathbf{n} \cdot (\mathbf{T}_{E,g} - \mathbf{T}_{E,l}) \mathbf{n}. \quad (28)$$

The first product uses the conducting limit as the basic model.

7 Capillarity and curvature

Surface tension produces a pressure jump proportional to mean curvature:

$$\Delta p_\gamma = \gamma \kappa. \quad (29)$$

For an axisymmetric graph $r = R(z)$, using the convention that the gas is toward increasing r , one convenient normal is

$$\mathbf{n} = \frac{(1, -R_z)}{\sqrt{1 + R_z^2}}. \quad (30)$$

The axisymmetric curvature is

$$\kappa = \frac{1}{R\sqrt{1 + R_z^2}} - \frac{R_{zz}}{(1 + R_z^2)^{3/2}}. \quad (31)$$

The first term is azimuthal curvature and the second is meridional curvature.

8 Young–Laplace–Maxwell balance

The quasi-static conducting-liquid normal stress balance is

$$\gamma \kappa = \Delta p + \frac{\varepsilon_g}{2} E_n^2, \quad (32)$$

where Δp is a constant pressure offset. Because curvature sign conventions vary, the implementation must choose one convention and maintain it consistently.

A useful residual is

$$\mathcal{R}(s) = \gamma \kappa(s) - \Delta p - \frac{\varepsilon_g}{2} E_n(s)^2. \quad (33)$$

A candidate equilibrium interface should satisfy

$$\mathcal{R}(s) \approx 0 \quad (34)$$

along Γ . The RMS residual is

$$\mathcal{R}_{\text{RMS}} = \left(\frac{1}{L_\Gamma} \int_\Gamma \mathcal{R}^2 ds \right)^{1/2}. \quad (35)$$

This residual is the bridge between theory and computation: it tells the code how physically balanced a guessed interface is.

9 Space-charge shielding theory

Poisson’s equation requires a closure for ρ_e . The project should use a hierarchy of closures rather than claiming a complete microscopic emission model.

9.1 Closure A: prescribed localized charge cloud

The simplest shielding model is

$$\rho_e(r, z) = \rho_0 \exp \left[-\frac{(r - r_a)^2 + (z - z_a)^2}{2\ell^2} \right], \quad (36)$$

where (r_a, z_a) is the apex location and ℓ is a shielding length. This model tests how a localized charge distribution changes the electric field.

9.2 Closure B: threshold-activated effective charge

A more physical effective closure activates charge only above a field scale E_c :

$$\rho_e = \rho_{\max} \left[1 - \exp \left(-\frac{|\mathbf{E}| - E_c}{E_s} \right) \right]_+, \quad (37)$$

where $[x]_+ = \max(x, 0)$, E_s smooths the transition, and $\rho_{\max}$ is a saturation density. Since ρ_e depends on $\mathbf{E} = -\nabla\phi$, this gives a nonlinear Poisson problem.

A fixed-point iteration is

$$\nabla^2 \phi^{(k+1)} = -\rho_e^{(k)} / \varepsilon_0, \quad (38)$$

$$\mathbf{E}^{(k+1)} = -\nabla \phi^{(k+1)}, \quad (39)$$

$$\tilde{\rho}_e^{(k+1)} = F(|\mathbf{E}^{(k+1)}|), \quad (40)$$

$$\rho_e^{(k+1)} = (1 - \omega) \rho_e^{(k)} + \omega \tilde{\rho}_e^{(k+1)}, \quad (41)$$

where $0 < \omega \leq 1$ is an under-relaxation parameter.

9.3 Later extension: drift-diffusion-Poisson

A more complete gas charge model solves

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \mathbf{J} = S, \quad \mathbf{J} = \mu_c \rho_e \mathbf{E} - D_c \nabla \rho_e, \quad (42)$$

coupled with Poisson's equation. This is a later extension because it introduces mobility, diffusion, and source parameters that may not be known in the first stage.

10 Leaky-dielectric extension

A real liquid may not be a perfect conductor. In a leaky-dielectric model,

$$\nabla \cdot (\varepsilon_g \nabla \phi_g) = -\rho_{e,g} \quad \text{in } \Omega_g, \quad (43)$$

$$\nabla \cdot (\varepsilon_l \nabla \phi_l) = -\rho_{e,l} \quad \text{in } \Omega_l. \quad (44)$$

If Ohmic conduction dominates inside the liquid,

$$\mathbf{J}_l = \sigma_l \mathbf{E}_l = -\sigma_l \nabla \phi_l, \quad (45)$$

and charge conservation gives

$$\nabla \cdot (\sigma_l \nabla \phi_l) = 0. \quad (46)$$

At the interface,

$$\phi_g = \phi_l, \quad (47)$$

$$\mathbf{n} \cdot (\varepsilon_g \mathbf{E}_g - \varepsilon_l \mathbf{E}_l) = q_s. \quad (48)$$

Surface charge conservation has the general form

$$\frac{\partial q_s}{\partial t} + \nabla_s \cdot (q_s \mathbf{u}_s) + q_s (\nabla_s \cdot \mathbf{u}_s) = \mathbf{n} \cdot (\sigma_l \mathbf{E}_l - \sigma_g \mathbf{E}_g) - j_{\text{emit}}. \quad (49)$$

This is included to show the continuum target, but the first working solver should not rely on solving this full system.

11 Nondimensionalization

Let L be a characteristic length scale, V_0 the voltage scale, and γ the surface tension. Define

$$\hat{r} = \frac{r}{L}, \quad \hat{z} = \frac{z}{L}, \quad \hat{\phi} = \frac{\phi}{V_0}. \quad (50)$$

Then

$$\mathbf{E} = -\frac{V_0}{L} \hat{\nabla} \hat{\phi}. \quad (51)$$

The dimensionless charge density is

$$\hat{\rho}_e = \frac{L^2 \rho_e}{\varepsilon_0 V_0}. \quad (52)$$

The electric-to-capillary stress ratio is

$$\text{Bo}_E = \frac{\varepsilon_0 V_0^2}{\gamma L}. \quad (53)$$

This parameter controls how strongly electric stress competes with surface tension.

12 Theoretical deliverables before coding is considered successful

Before the solver is treated as a working theoretical product, it must demonstrate:

1. **Taylor limit:** recovery of $\alpha_T \approx 49.3^\circ$ in the zero-space-charge conducting limit.
2. **Correct operator:** implementation of the axisymmetric Laplace/Poisson operator, including axis regularity.
3. **Stress consistency:** Maxwell pressure scales as V_0^2 .
4. **Curvature consistency:** curvature signs and magnitudes are stable for known curves.
5. **Residual meaning:** equilibrium-like shapes have smaller $\mathcal{R}_{\text{RMS}}$ than deliberately poor shapes.
6. **Space-charge effect:** the Poisson model changes peak field and can represent shielding relative to the Laplace case.
7. **Parameter transparency:** every non-ideal parameter, such as E_c , ρ_{max} , or ℓ , is declared as a model parameter rather than hidden.

13 Conclusion

The theory required for the working product is now defined. The solver should begin from the classical Taylor cone, solve axisymmetric Laplace/Poisson electrostatics, compute electric fields, evaluate Maxwell pressure and capillary curvature on an interface, and use the Young–Laplace–Maxwell residual as the central physical diagnostic. Space-charge shielding should be introduced as an explicit Poisson closure, first phenomenologically and then through threshold-activated nonlinear iteration. This gives a rigorous theoretical foundation for the separate coding implementation outline.

References

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